How CareerGuide AI Works

A comprehensive technical overview of the AI-powered career counseling system — architecture, data, and training.

📘 Technical Overview Document

🤔 What is CareerGuide AI?

An overview of what this system does and the problem it solves.

An offline AI-powered career counseling system

CareerGuide AI is a locally-deployed application that conducts structured, conversational career assessments and returns personalized graduation course and career path recommendations. It combines a skills-matching recommendation engine with a locally-running large language model (LLM) to produce natural, context-aware guidance — without any cloud dependency.

💡 **CORE PURPOSE**

The system replicates the function of a human career counselor — gathering information about a student's academic background, interests, and goals, then mapping those inputs to relevant course-career pathways using structured data and AI-generated narrative explanation.

💻	🎯	📡	🔒
40+	**30+**	**0**	**100%**
Graduation Courses	Career Paths	Internet Needed	Private & Offline

🗺️ System Architecture

How all components of the application interact at runtime:

```
[ YOU — typing in the chat ]
            ↓ your message
[ FLASK SERVER — the middleman running on your laptop ]
            ↓ looks up matching careers
[ DATASET — list of courses, careers, salaries ]
            ↓ passes results to AI
[ OLLAMA AI — the brain, thinks and writes a reply ]
            ↓ sends reply back
[ YOU — reading the AI's personalized advice ]
```

🔑 **Key idea:** Nothing leaves your laptop. The AI thinks, the data is stored, and the reply is shown — all on your own computer. No internet needed after setup!

🧩 The 3 Core Components

The application is composed of three integrated layers:

🌐 Component 1 — Web Layer (Flask)

Flask is a lightweight Python web framework that serves the application's frontend (HTML pages) and exposes REST API endpoints for the chat interface, profile updates, and recommendation queries. It handles session management via server-side filesystem sessions and routes all requests between the browser and the backend logic.

📊 Component 2 — Data Layer (CSV Datasets)

Three structured CSV files stored locally contain the domain knowledge: course definitions, career profiles with salary and growth data, and academic field taxonomies. At startup, all data is loaded into memory as Python dictionaries for fast lookup. The recommendation engine performs keyword-overlap scoring against this dataset at query time.

🤖 Component 3 — AI Layer (Ollama + Llama 3.2)

Ollama is an open-source runtime that manages local LLM inference. It exposes a local REST API at `localhost:11434` that the Flask backend calls with each user message. The Llama 3.2 3B model processes the full conversation history plus injected recommendation context and returns a natural language response. All inference runs on the local CPU — no external API calls are made.

👣 Request Lifecycle — Step by Step

The complete flow from user input to AI response:

1

User submits a message via the chat interface

The browser sends a POST request to `/api/chat` with the message payload in JSON format. The user's profile data (interests, field, background) stored in the sidebar form is also synced to the server session via `/api/update-profile`.

2

Flask API endpoint processes the request

The `/api/chat` route handler retrieves the current session (conversation history + user profile), appends the new user message to the history, and passes the full context to the recommendation engine and LLM pipeline.

3

Recommendation engine scores and ranks careers

The user's interests are tokenized and matched against each career's `required_skills` field using substring overlap scoring. Careers are ranked by match count and the top 5 are selected. For each matched career, its linked course IDs are resolved to full course records from the courses dataset.

4

Recommendation context is injected into the system prompt

The top 5 careers, their salaries, growth ratings, and recommended courses are serialized into a structured text block and appended to the LLM's system prompt. This grounds the model's response in factual, domain-specific data rather than relying solely on its pre-trained knowledge.

5

Llama 3.2 generates a response via Ollama inference

The full message array (system prompt + conversation history) is sent to Ollama's `/api/chat` endpoint at `localhost:11434`. The model runs inference locally at temperature 0.7 with a 512-token output limit and returns a structured JSON response containing the assistant's message.

6

Response is returned and conversation history updated

The assistant's reply is appended to the session's message history and returned to the browser as JSON. The frontend renders it in the chat UI. On the next message, the full updated history is sent again — enabling multi-turn contextual conversation throughout the session.

🎯 Recommendation Algorithm — Detail

How the skills-interest matching engine scores and ranks career paths:

Keyword Overlap Scoring (Bag-of-Words Matching)

Each career record contains a comma-separated `required_skills` field. The user's stated interests are split into tokens and each token is tested for substring containment against each skill token in every career record. The score for a career equals the count of interest tokens that match at least one skill. Careers are sorted descending by score; top 5 are returned.

💡 **SCORING EXAMPLE**

User input interests: **"Python, data, math"**

Scoring against careers:
→ **Data Scientist** — skills: Python, ML, Statistics, SQL → *2 matches (Python, data/Statistics)* → *Score: 2*
→ **Web Developer** — skills: HTML, CSS, JavaScript → *0 matches* → *Score: 0*
→ **Business Analyst** — skills: Excel, SQL, Statistics, Communication → *1 match (Statistics≈math)* → *Score: 1*

Result: Data Scientist ranked first with the highest overlap score.

Your Interest	Matched Career	Match Score	Avg Salary
Python, data, math	Data Scientist	⭐⭐⭐ High	₹12,00,000/yr
coding, problem solving	Software Developer	⭐⭐⭐ High	₹8,00,000/yr
design, creativity	UX Designer	⭐⭐ Medium	₹7,50,000/yr
people, helping others	Clinical Psychologist	⭐⭐ Medium	₹5,00,000/yr
finance, numbers	Financial Analyst	⭐⭐⭐ High	₹9,00,000/yr

🧠 The AI Model — Llama 3.2 by Meta

Technical profile of the language model powering the chat interface:

Model: Llama 3.2 3B (Meta AI, 2024)

Llama 3.2 is Meta's open-weight large language model released in September 2024. The 3B variant has **3.21 billion parameters**, uses a transformer decoder architecture with grouped query attention (GQA), and supports a 128,000-token context window. It was trained using supervised fine-tuning (SFT) and reinforcement learning from human feedback (RLHF).

Specification	Value
Model family	Llama 3.2
Developer	Meta AI
Release date	September 2024
Parameter count	3.21 billion
Architecture	Transformer Decoder (auto-regressive)
Attention mechanism	Grouped Query Attention (GQA)
Context window	128,000 tokens
Vocabulary size	128,256 tokens (tiktoken BPE)
Training precision	BFloat16
Quantized size (Q4)	~2.0 GB on disk
License	Llama 3.2 Community License (free for most uses)
Inference runtime	Ollama (CPU and GPU supported)

Why does it run offline after download?

During training, the model's learned knowledge is encoded as billions of numerical weights stored in a file (~2GB quantized). Once downloaded to your laptop, Ollama loads these weights into RAM and runs the model's inference computations locally on the CPU. No internet connection is needed because all the model's knowledge is embedded in those weight files — the model does not "look up" answers online.

📚 Llama 3.2 Training Data — What It Was Trained On

The model's knowledge comes from a massive pre-training corpus. Below is a breakdown of the publicly disclosed training data sources, volume, and composition:

Total Pre-Training Data Volume

Llama 3.2 3B was trained on approximately **9 trillion tokens** of text data — equivalent to roughly 6.75 trillion words, or the content of approximately **45 million books**. This is the total corpus used across the Llama 3.x training runs; the 3B model uses a curated subset of this data optimized for the smaller model size.

📄
9T
Tokens trained on

📚
~45M
Book equivalents

🌐
50+
Languages covered

📅
2024
Knowledge cutoff

Data Source Category	Description	Approx. Share
Web Crawl (CommonCrawl)	Filtered snapshots of billions of public web pages — news sites, blogs, forums, educational content, documentation. Heavily filtered for quality using classifier-based deduplication and heuristic cleaning.	~65–70%
Wikipedia	Full text of Wikipedia in multiple languages (primarily English). Covers encyclopedic knowledge across science, history, geography, technology, medicine, law, culture, and more.	~4–5%
Books / Long-Form Text	Digitized books including fiction, non-fiction, academic textbooks, and reference works. Sources include Project Gutenberg (70,000+ public domain books), and other curated book corpora. Contributes deep reasoning and narrative understanding.	~4–5%
Academic Papers (ArXiv, PubMed, Semantic Scholar)	Peer-reviewed scientific papers across computer science, mathematics, physics, biology, chemistry, medicine, and engineering. Over 2 million papers from ArXiv alone; PubMed covers 35+ million biomedical citations.	~2–3%
Code Repositories (GitHub, Stack Overflow)	Open-source code from GitHub across 40+ programming languages (Python, Java, C++, JavaScript, Go, Rust, etc.) and developer Q&A from Stack Overflow. Enables code generation and technical reasoning.	~8–10%
News Articles	Curated news content from thousands of publications worldwide. Covers current events, business, science, politics, and international affairs up to the training cutoff date.	~3–4%

Data Source Category	Description	Approx. Share
Multilingual Web Text	Non-English web content covering 50+ languages including German, French, Spanish, Italian, Portuguese, Hindi, Japanese, Korean, Chinese, Arabic, and more.	~5–8%
Synthetic Data (Meta-generated)	High-quality synthetic conversations, reasoning chains, and instruction-following examples generated using larger Llama models (Llama 3.1 405B). Used specifically for fine-tuning alignment, helpfulness, and safety behaviour.	Undisclosed

📌 **Note on transparency:** Meta has not publicly released the exact dataset composition percentages for Llama 3.2. The shares above are estimates based on Meta's published Llama 3 technical report, the Llama 3.2 model card, and independent analysis by the research community. The broad categories and sources are confirmed; the exact proportions are approximate.

📖 Specific Notable Training Sources

Key named datasets and corpora that are publicly confirmed or widely reported as part of the Llama 3.x training pipeline:

Dataset / Source	Type	Scale	What It Contributed
CommonCrawl (CC-Main)	Web crawl	Petabytes raw; filtered to ~trillions of tokens	Broad world knowledge, language diversity, factual content
Wikipedia (EN + multilingual)	Encyclopedia	~20GB text; 6.7M+ English articles	Factual accuracy, structured knowledge, definitions
Project Gutenberg	Books	70,000+ books, ~60GB	Literary language, long-form reasoning, pre-1927 published works
Books3 / The Pile	Books corpus	~100GB, ~200,000 books	Fiction and non-fiction narrative, academic writing style
ArXiv	Academic papers	2M+ papers, ~270GB	Scientific and mathematical reasoning, technical terminology
PubMed Central	Biomedical papers	4M+ full-text articles	Medical and biological knowledge
GitHub (open-source code)	Source code	Hundreds of GB across 40+ languages	Code generation, debugging, software concepts
Stack Overflow / Stack Exchange	Q&A forums	Millions of Q&A pairs	Technical problem-solving, developer knowledge
C4 (Colossal Clean Crawled Corpus)	Filtered web	~750GB	Clean English web text, general knowledge

Dataset / Source	Type	Scale	What It Contributed
OpenWebText	Web text	~38GB	High-quality web content (Reddit-upvoted links)
RefinedWeb	Filtered web crawl	~5TB	Quality-filtered CommonCrawl derivative
DM Mathematics	Math problems	Millions of problems	Mathematical reasoning, algebra, calculus
Europarl	Parliamentary records	~60M sentences, 21 EU languages	Multilingual alignment, formal language
CC-News	News articles	~76GB, 63M articles	Current events knowledge, journalistic writing
Synthetic SFT data (Meta)	AI-generated	Millions of examples	Instruction following, conversation, safety alignment

💡 **SCALE IN PERSPECTIVE**

9 trillion tokens is roughly equivalent to reading every English Wikipedia article **6,000 times over**, or reading a new book every minute for **85 years without stopping**. This breadth of training is why the model can discuss careers, salary trends, educational pathways, and give coherent advice across dozens of fields.

What this means for career counseling quality

Because the model was trained on academic papers, Wikipedia, educational content, career forums, news, and professional Q&A sites, it has broad background knowledge on occupations, industries, educational requirements, salary trends, and skill demands. The app harnesses this knowledge and grounds it further with the local CSV dataset to produce recommendations that are both conversationally natural and factually anchored.

📁 The Data — Where Does the Information Come From?

The app uses three simple spreadsheet files saved on your laptop.

File Name	What's Inside	Example
courses.csv	40+ graduation courses with skills taught and description	Machine Learning, Data Science, Web Development...
careers.csv	30+ career paths with salary, job growth, and required skills	Data Scientist — ₹12L/yr — Very High growth
fields.csv	8 academic fields and what careers they lead to	Computer Science, Business, Healthcare, Arts...

📝 **You can add more data!** These files are just spreadsheets. Open them in Excel or Notepad, add a new row following the same format, and the app will automatically include your new course or career next time it starts.

🔒 Privacy — Is My Data Safe?

Yes — nothing leaves your laptop. Ever.

Unlike apps like ChatGPT or Google, CareerGuide AI does NOT send your messages to any server on the internet. Everything happens locally: the AI thinks on your CPU, the data is on your hard drive, and the replies stay on your screen. No company can see your conversations.

📤 **0** Data sent to internet

🔑 **0** API keys needed

👤 **0** Accounts required

☁️ **0** Cloud storage used

🛠️ Technologies Used

Here are the tools that make CareerGuide AI work:

Technology	What It Is	What It Does in This App
Python	A popular programming language	Runs the entire backend of the app
Flask	A Python web framework	Creates the website you see in the browser
Ollama	A free tool to run AI models locally	Loads and runs the Llama AI on your laptop

Technology	What It Is	What It Does in This App
Llama 3.2	An AI language model by Meta	Understands your messages and writes replies
CSV Files	Simple spreadsheet format	Stores all course and career data
HTML / CSS / JS	Web page languages	Makes the chat interface look good and work

✅ Quick Quiz — Test Your Understanding

Let's see if you understood how the app works! Click "Show Answer" to reveal each answer.

🧠 Answer these questions:

Q1. What does Flask do in this app?

Show Answer

Q2. Why doesn't the app need the internet after setup?

Show Answer

Q3. How does the app decide which career matches you?

Show Answer

Q4. What is Ollama?

Show Answer

Q5. Can you add new careers to the app? How?

Show Answer

📖 Glossary — Key Words Explained

AI (Artificial Intelligence)	Computer programs that can do tasks that normally require human intelligence, like understanding language or making decisions.
Language Model	A type of AI trained on huge amounts of text. It learns patterns in language and can generate human-like responses to questions.
Flask	A Python library used to build websites and web applications. It handles the connection between your browser and the Python code running on your laptop.
Ollama	A free, open-source tool that lets you run large AI models locally on your own computer without needing the internet or paying for API access.
Llama 3.2	An AI language model created by Meta (Facebook's parent company) and released for free. The "3B" version has 3 billion parameters and runs on regular laptops.

CSV File	A simple spreadsheet file (Comma Separated Values). Each line is a row of data, with columns separated by commas. Can be opened in Excel.
Localhost	A special address (http://localhost:5000) that means "this computer". When you visit this address, you're connecting to a server running on your own laptop.
Recommendation Engine	The part of the app that matches your interests with careers by comparing and scoring them. Similar to how Netflix recommends movies based on what you've watched.
Parameters (in AI)	Numbers inside an AI model that were adjusted during training. More parameters generally means a smarter model. Llama 3.2 3B has 3 billion parameters.

🌟 Summary

In one sentence: CareerGuide AI is an offline chat application that uses a local AI model (Llama 3.2) and a career database (CSV files) to recommend personalized graduation courses and career paths based on your interests — all running privately on your own laptop using Python and Flask.

🚀 What makes this app special?

Most AI tools like ChatGPT send your data to a server in another country. This app runs **everything on your own laptop** — your conversations are private, it works without internet, and it costs nothing to run after setup. That's the power of open-source AI combined with thoughtful software design.

CareerGuide AI — Offline Career Counselor for Graduation Course Recommendations

Document prepared for Grade 9 students | All concepts explained in plain language